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Replication Project Proposal: Large Language Models and Labor Market Signals

Paper: "Labor Market Signals: The Role of Large Language Models," by Kian Abbas Nejad, Giuseppe Musillo, Till Wicker, and Niccolò Zaccaria. *Journal of Labor Economics*, published online October 2026. https://doi.org/10.1086/744669

Replication package: https://doi.org/10.7910/DVN/LPIREO (Harvard Dataverse). I downloaded it and confirmed it includes the analysis data (.dta) and the code for every regression table. Only the raw cover-letter texts and browser histories are withheld (IRB/GDPR), and none of the tables below depend on them.

Language: Stata (tables), with MATLAB for the structural model. My replication will be in R.

Research question: The effect of job applicants' access to ChatGPT on the quality of their cover letters and how recruiters rate them, i.e., whether a cover letter still signals applicant quality once writing is nearly free.

Why this paper: As a CS undergrad who follows how AI agents are spreading through everyday life and the workforce, I want to see econometric evidence of what that integration actually does to labor markets. This paper addresses a big issue, which is that if anyone can produce a polished cover letter for free, employers lose a signal they use to screen applicants. It also connects directly to my own work outside the MSQE, where I'm building an LLM and automation system to optimize my job search.

Summary statistics: Online Appendix Table A1, "Balance Table: Demographics," all columns (applicant characteristics by control and ChatGPT groups, with tests of differences).

Main OLS result: Table 1, "Effect of ChatGPT on Cover Letter," columns (1)–(6). Standardized recruiter scores are regressed on the ChatGPT treatment dummy with controls, recruiter-by-firm and school fixed effects, and standard errors clustered by applicant. Column (1): 0.222 SD increase in the overall score (SE 0.117, N = 274).